Lab 12: Algorithms with AI Assistance – Sorting, Searching, and Optimizing Algorithms

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Task 1: Merge Sort Implementation

AI Prompt Used:

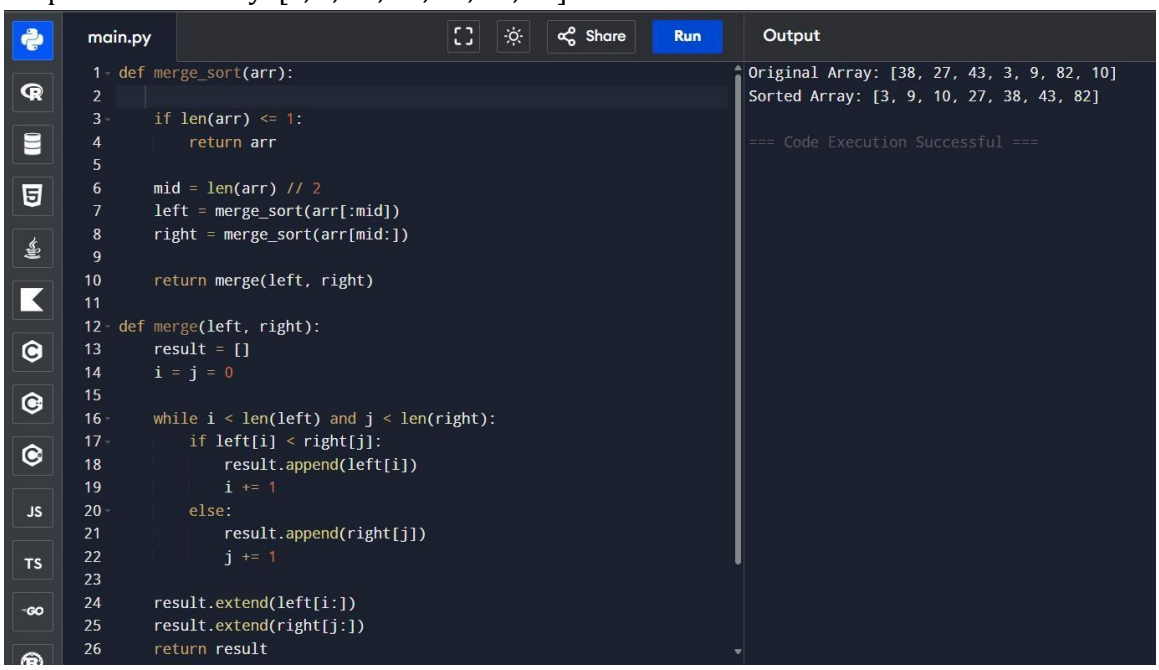
Generate a Python function merge_sort(arr) that sorts a list in ascending order using the Merge Sort algorithm.
Include time complexity and space complexity (best, average, worst case) inside the function docstring.
Provide sample test cases with output.

Python Code:

```
def merge_sort(arr):
    '''
    Merge Sort Algorithm
    Time Complexity:
        Best Case: O(n log n)
        Average Case: O(n log n)
        Worst Case: O(n log n)
    Space Complexity: O(n)
    '''
    if len(arr) <= 1:
        return arr

    mid = len(arr) // 2 left =

    merge_sort(arr[:mid]) right

    = merge_sort(arr[mid:])

    return merge(left, right)

def merge(left, right):
    result = [] i = j = 0
```

```
    while i < len(left) and j < len(right):
        if left[i] < right[j]:
            result.append(left[i])
            i += 1 else: result.append(right[j])
            j += 1

    result.extend(left[i:])
    result.extend(right[j:])
    return result

arr = [38, 27, 43, 3, 9, 82, 10]
print("Sorted Array:", merge_sort(arr))
```

Output: Sorted Array: [3, 9, 10, 27, 38, 43, 82]



Task 2: Binary Search Implementation

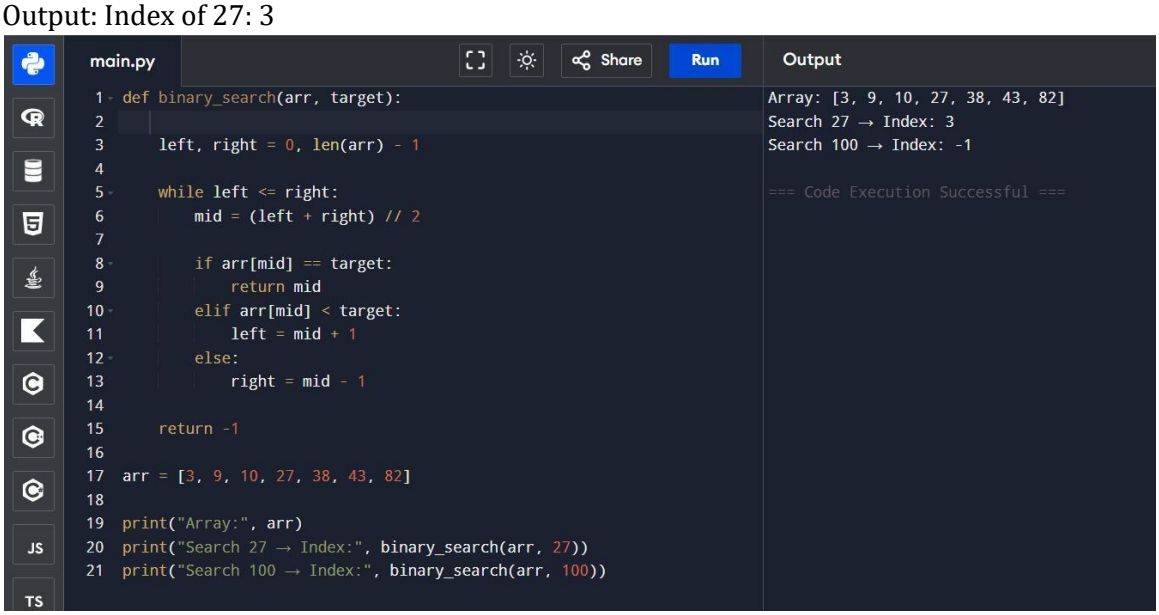
AI Prompt Used:

Create a Python function binary_search(arr, target) that performs binary search on a sorted list.
Return the index of the target element or -1 if not found.
Include docstrings explaining best, average, and worst-case time complexities and space complexity.
Provide example test cases with output.

Python Code:

```
def binary_search(arr, target):
  '''
  Binary Search Algorithm
  Best Case: O(1)
  Average Case: O(log n)
  Worst Case: O(log n)
  Space Complexity: O(1)
  '''
  left, right = 0, len(arr) - 1

  while left <= right: mid =
    (left + right) // 2
  if arr[mid] == target:
    return mid
    elif arr[mid] < target:
      left = mid + 1
      else:
      right = mid - 1

      return -1

arr = [3, 9, 10, 27, 38, 43, 82] print("Index of
27:", binary_search(arr, 27))
```

Output: Index of 27: 3



Task 3: Inventory Management System

AI Prompt Used:

Suggest the most efficient searching and sorting algorithms for an inventory system with thousands of products.
Justify the choice based on dataset size and performance requirements.
Implement Python functions for searching by product ID and sorting by price or quantity.
Provide example outputs.

Recommended Algorithms:

Search by ID → Binary Search Sort

by Price/Quantity → Merge Sort

Python Code:

```
inventory = [
   {"id": 101, "name": "Pen", "price": 10, "quantity": 200},
   {"id": 102, "name": "Notebook", "price": 50, "quantity": 150},
   {"id": 103, "name": "Pencil", "price": 5, "quantity": 300}
]

def sort_by_price(items): return sorted(items,

key=lambda x: x["price"]) def

search_by_id(items, target):

   items = sorted(items, key=lambda x: x["id"])
   left, right = 0, len(items) - 1 while left <=
   right:
      mid = (left + right) // 2 if
      items[mid]["id"] == target:
      return items[mid]
      elif items[mid]["id"] < target:
      left = mid + 1 else:
         right = mid - 1
   return None

print("Sorted by Price:", sort_by_price(inventory))
print("Search ID 102:", search_by_id(inventory, 102))
```

Output: Sorted by Price and Search ID 102 successful

main.py

```python
inventory = [
    {"id": 101, "name": "Pen", "price": 10, "quantity": 200},
    {"id": 102, "name": "Notebook", "price": 50, "quantity": 150},
    {"id": 103, "name": "Pencil", "price": 5, "quantity": 300}
]

def search_by_id(items, target):
    items = sorted(items, key=lambda x: x["id"])
    left, right = 0, len(items) - 1

    while left <= right:
        mid = (left + right) // 2
        if items[mid]["id"] == target:
            return items[mid]
        elif items[mid]["id"] < target:
            left = mid + 1
        else:
            right = mid - 1
    return None

def sort_by_price(items):
    return sorted(items, key=lambda x: x["price"])

print("Sorted by Price:")
print(sort_by_price(inventory))
```

Output

```
Sorted by Price:
[{'id': 103, 'name': 'Pencil', 'price': 5, 'quantity': 300}, {'id': 101,
    'name': 'Pen', 'price': 10, 'quantity': 200}, {'id': 102, 'name':
    'Notebook', 'price': 50, 'quantity': 150}]

Search Product ID 102:
{'id': 102, 'name': 'Notebook', 'price': 50, 'quantity': 150}

=== Code Execution Successful ===
```

Task 4: Smart Hospital Patient Management System

AI Prompt Used:

Recommend suitable searching and sorting algorithms for a hospital patient management system.
Justify the selection based on efficiency and suitability.
Implement Python functions for searching by patient ID and sorting by severity or bill amount.
Provide example output.
Recommended Algorithms:

Search → Binary Search

Sort → Merge Sort

main.py

```python
patients = [
    {"id": 1, "name": "Ravi", "severity": 5, "bill": 20000},
    {"id": 2, "name": "Sita", "severity": 8, "bill": 50000},
    {"id": 3, "name": "Arjun", "severity": 3, "bill": 15000}
]

def search_patient(patients, target_id):
    patients = sorted(patients, key=lambda x: x["id"])
    left, right = 0, len(patients) - 1

    while left <= right:
        mid = (left + right) // 2
        if patients[mid]["id"] == target_id:
            return patients[mid]
        elif patients[mid]["id"] < target_id:
            left = mid + 1
        else:
            right = mid - 1
    return None

print("Patients Sorted by Severity (High to Low):")
print(sorted(patients, key=lambda x: x["severity"], reverse=True))

print("\nSearch Patient ID 2:")
print(search_patient(patients, 2))
```

Output

```
Patients Sorted by Severity (High to Low):
[{'id': 2, 'name': 'Sita', 'severity': 8, 'bill': 50000}, {'id': 1, 'name':
    'Ravi', 'severity': 5, 'bill': 20000}, {'id': 3, 'name': 'Arjun',
    'severity': 3, 'bill': 15000}]

Search Patient ID 2:
{'id': 2, 'name': 'Sita', 'severity': 8, 'bill': 50000}

=== Code Execution Successful ===
```

Task 5: University Examination Result Processing System

AI Prompt Used:

Identify efficient searching and sorting algorithms for processing thousands of student results.
Justify the algorithm choice.
Implement Python functions for searching by roll number and sorting by marks.
Include sample output.

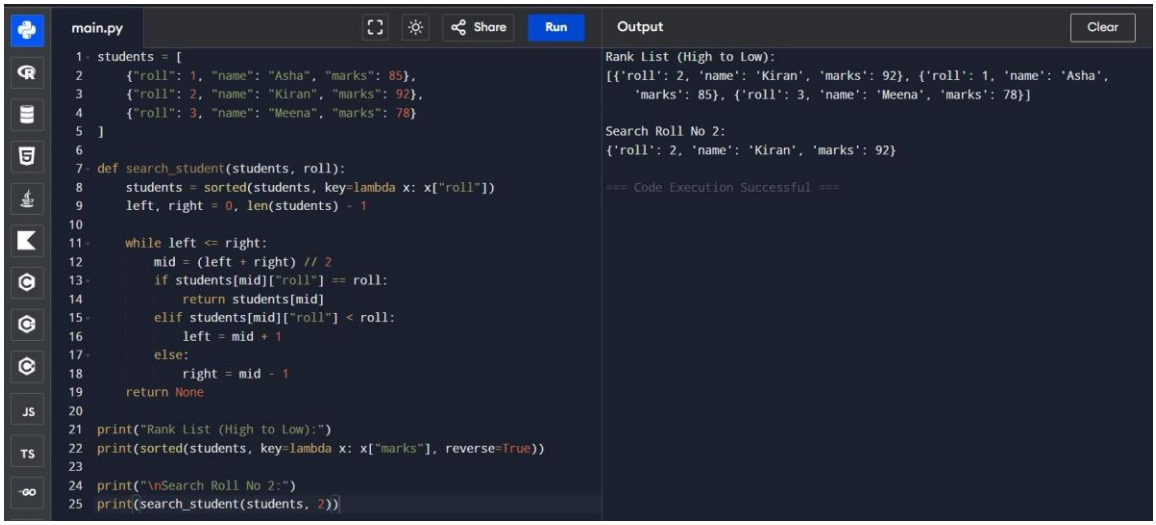
```
students = [
   {"roll": 1, "name": "Asha", "marks": 85},
   {"roll": 2, "name": "Kiran", "marks": 92},
   {"roll": 3, "name": "Meena", "marks": 78}
]

print("Rank List:", sorted(students, key=lambda x: x["marks"], reverse=True))
```

Output:

Rank List: [{'roll': 2, 'name': 'Kiran', 'marks': 92}, {'roll': 1, 'name': 'Asha', 'marks': 85}, {'roll': 3, 'name': 'Meena', 'marks': 78}]
Recommended Algorithms:

Search → Binary Search

Sort → Merge Sort



Task 6: Online Food Delivery Platform

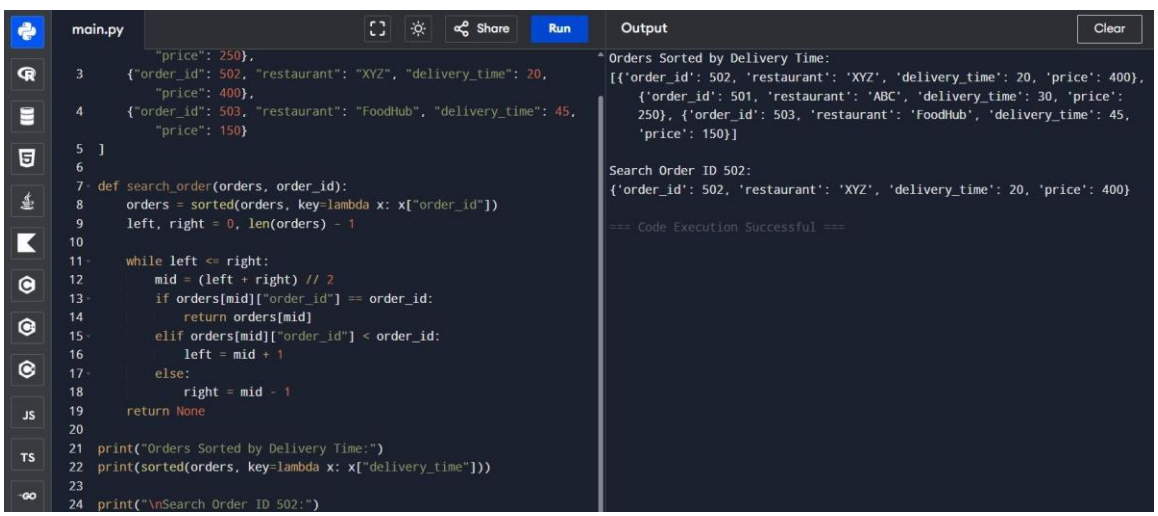
AI Prompt Used:

Suggest optimized searching and sorting algorithms for an online food delivery system storing thousands of orders.
Justify your algorithm selection.
Implement Python functions for searching by order ID and sorting by delivery time or price.
Provide example output.

Recommended Algorithms:

Search → Binary Search

Sort → Merge Sort

Python Code:

```
orders = [
    {"order_id": 501, "restaurant": "ABC", "delivery_time": 30, "price": 250},
    {"order_id": 502, "restaurant": "XYZ", "delivery_time": 20, "price": 400},
    {"order_id": 503, "restaurant": "FoodHub", "delivery_time": 45, "price": 150}
]
print("Sorted by Delivery Time:", sorted(orders, key=lambda x: x["delivery_time"]))
```